Leaf Stomata Microscope Detective

Count stomata and rank leaves by water strategy

At a glance: Count the breathing pores and rank leaves by how they manage water. Plan about 80 minutes for the core block; 4 teams of 4 students, one digital microscope per team.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: stomata, water loss, gas exchange, microscopy, data sampling (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- leaf samples: 3 to 4 fresh leaves per team from contrasting habitats (about 24 total, collect on campus the morning of)
- clear tape: clear packing/transparent tape, a few short strips per team (2 rolls for the camp)
- clear nail polish: 1 small bottle covers all 4 teams (bring 2; ventilated station only)
- glass slides: about 16 per run; a 50 to 72 slide box for the camp
- prepared stomata slides: about 4 as a backup for failed peels (order a small set or borrow)
- digital microscopes: one per team (4; two on hand, rotate or borrow two more); screen models need charging, USB-cable models need a laptop/tablet
- grid/counting sheets: 1 per team or per student (20 with buffer); counting sheet in this guide
- gloves: latex-free, a box at the polish station (optional, not class-wide)

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

02 Stage the scoring sheet, the grid/counting sheets, and 4 backup prepared stomata slides, and decide the counting convention (count every stoma fully inside the field, at least three fields per leaf) so all teams count the same way.

03 Set up one ventilated polish-and-dry station with the nail polish, a box of latex-free gloves, and a drying tray, away from the microscope tables. No goggles or containment trays are needed: this is a dry counting activity with one small ventilated polish step.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Count the breathing pores and rank leaves by how they manage water." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Make impressions and count

0:18 - 0:44

Teams make clear-polish impressions of 3 to 4 leaves at the ventilated station, lift them with tape onto labeled slides, then focus each slide and count stomata in at least three fields per leaf. Circulate, check that they count the same way every time and log every field, and resist counting for them.

Phase 5 Average and rank (within the work block)

Teams average their counts per leaf and rank the leaves from water-saving (fewer stomata) to water-spending (more stomata), citing the numbers.

Phase 6 Check a second surface or species (extension)

Teams who finish make an impression of a top surface or a different species and compare it to their lower-surface counts, noting where most stomata are.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, defend their ranking with their counts, then reset the station: bin used tape, peels, and browned leaves, return microscopes and reusable slides to the bin, cap the polish, wipe the tables, and wash hands (remove gloves first if worn).

TROUBLESHOOTING

Problem: A team's counts vary wildly between fields. Fix: Standardize the count: same magnification, count only stomata fully inside the field, do not move the stage mid-count, and average at least three fields; treat it as a data-quality coaching moment.

Problem: The peel shows nothing under the microscope. Fix: Confirm the polish dried fully before taping, that the underside was painted (most stomata are there), and that the leaf was not browned by the solvent; if it failed, switch that team to a backup prepared slide so they can still count and rank.

Problem: Stomata are hard to find or out of focus. Fix: Start at low magnification to locate the impression, then increase magnification and re-focus; make sure the taped peel is flat with no bubbles, and re-make the peel if it is smeared.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to stomata: pores for gas exchange.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why count several fields instead of one?
- What does a high stomata count suggest about a leaf's habitat?
- Where might error sneak into your count?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Slide or peel quality	25
Counting accuracy	30
Ranking evidence	25
Team data table	10
Cleanup and care	10
Total	**100**

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES California Academy of Sciences, Stomata Printing microscope investigation (clear nail polish + clear tape impressions, grades 3 to 12); Rothamsted Bioimaging, Measuring Stomatal Density using nail varnish; Science and Plants for Schools (SAPS), Measuring Stomatal Density.